AUTHENTICLENS

KNOWLEDGE BASE - DOCUMENT 2 OF 9

DOCUMENT 2 OF 9 - PHOTOGRAPHIC PHYSICS AND IMPERFECTIONS

Corrected technical reference for optics, sensors, computational photography, compression, and realistic imperfections

Version 2.0 - April 2026. Fully revised and corrected from the original v1.0 document. This version integrates the audit findings, errata, Lightroom/editorial separation, mobile/compression caveats, provenance limits, and Gemini / Nano Banana / Nano Banana Pro execution requirements.

Prepared for: AuthenticLens - The Realism Architect
Target use: Gemini Gem knowledge-base upload
Primary image workflow: Gemini Pro reasoning + Nano Banana / Nano Banana Pro image generation and editing
Language: English (Canada)

Purpose

This document is the corrected photographic physics reference for AuthenticLens. It explains the real camera, lens, sensor, film, phone-camera, compression, colour-management, and lighting phenomena that must guide both authenticity analysis and realism enhancement. It replaces rigid “real photos always show X” assumptions with context-aware operational rules.

Intended Gemini Behaviour

- Use physics to explain why an image looks real, edited, AI-generated, or composited.
- Do not expect every real image to show obvious lens aberrations, clipping, noise, or grain; modern cameras and software correct many artifacts.
- Understand that smartphone images often show multi-frame HDR, noise reduction, sharpening, tone mapping, semantic segmentation, and artificial bokeh.
- When improving realism, add imperfections subtly and coherently. Never spray generic grain, blur, chromatic aberration, or vignette across every image.
- For product and e-commerce work, prioritize accurate colour, controlled sharpness, fabric visibility, and clean delivery over fake “gritty realism”.

Version 2.0 Corrections Applied

Old weakness	Updated correction
Natural vignetting formula rendered incorrectly.	Correct formula: I(theta) = I(0) x cos^4(theta), where theta is the off-axis angle.
Absence of chromatic aberration, vignetting, clipping, or noise treated as suspicious.	Modern lenses, RAW processors, phone pipelines, and image editors correct or suppress these traits. Absence alone is not AI evidence.
DSLR/film pipeline overrepresented.	Added computational photography, phone-camera processing, JPEG/social compression, colour delivery, and provenance caveats.
Physics was theoretical but not operational.	Each section now includes analysis instructions and editing/prompting instructions.

Section 1 - Optical Aberrations: Use Carefully

Real lenses can produce chromatic aberration, distortion, vignetting, flare, diffraction, field curvature, coma, and astigmatism. However, modern lens corrections and computational photography can reduce or remove these artifacts. Their presence supports photographic capture; their absence is not proof of AI.

Phenomenon	Physical meaning	Analysis use	Editing/prompting use
Chromatic aberration	Different wavelengths focus or magnify differently, creating colour fringing near high-contrast edges.	Look for coherent fringing near edges, especially frame periphery. Be cautious: software often corrects CA.	Add only subtle edge-specific fringing, not rainbow halos everywhere.

Vignetting	Brightness/saturation falloff toward image edges. Natural vignetting can follow cos^4 falloff; mechanical vignetting can be harder-edged.	Real vignetting depends on lens/aperture/sensor. A uniform Instagram-style vignette is editing, not AI.	Use mild corner falloff only when it matches lens/scene. Avoid heavy dark rings.
Lens distortion	Straight lines curve because of lens geometry: barrel, pincushion, or mustache distortion.	Wide-angle street/interior shots may show residual distortion or software correction. Perfect lines can be corrected real photos.	For phone/wide scenes, allow slight barrel distortion and edge stretching.
Lens flare and veiling glare	Internal reflections/scatter reduce contrast or create flare artifacts around bright sources.	Flare should relate to bright source and optical axis. Decorative flare unrelated to source is suspicious.	Add flare only when a visible or implied light source supports it.
Diffraction	Small apertures spread light into Airy patterns and reduce sharpness.	Uniform softness at small apertures is plausible in landscapes/product macro. It is not the same as missed focus.	Do not add diffraction unless a small-aperture look is intended.

Operational rule: optical imperfections must be spatially coherent. Random chromatic aberration, random flare, and generic vignette can make an image look more artificial.

Section 2 - Sensor Noise, Film Grain, and Digital Cleanliness

Real photographs contain sensor or film structure, but that structure can be hidden by denoising, resizing, compression, sharpening, and social media processing. AuthenticLens must distinguish sensor/film artifacts from AI texture hallucination.

Artifact	Confirmed / common behaviour	False-positive trap	Realism edit rule
Shot noise	Photon noise is random and increases relative visibility in low light and shadows.	AI can mimic noise; compression can destroy it; denoising can remove it.	Apply fine, random, luminance-dominant noise in shadows/midtones only when needed.
Read noise / banding	Sensor electronics can add structured noise in shadows or long exposures.	Banding can also come from compression or display capture.	Rarely add banding; use only for low-light phone/CCTV looks.
Colour noise	High ISO shadows can show chroma speckles; Lightroom can reduce	Denoising can smear colour edges and create plastic surfaces.	Prefer subtle luminance grain over strong coloured speckles.

	them.		
Film grain	Film grain is tonal and clumpy, often most visible in midtones; grain size depends on stock and ISO.	Uniform overlay grain across pure black/white and smooth skies looks fake.	Use grain tied to tone and image scale, not a generic texture layer.
AI texture noise	Noise-like pattern may repeat, correlate across surfaces, or ignore material boundaries.	Social compression can create blocks/ringing that look synthetic.	Diagnose before editing; do not mask every issue with grain.

Section 3 - Computational Photography and Phone-Camera Realism

Most everyday images are not single-exposure DSLR/film photographs. They are computational outputs: multi-frame capture, HDR merging, sharpening, denoising, segmentation, tone mapping, face/skin processing, and sometimes artificial depth effects.

Phone-camera trait	How it appears	Analysis implication	Prompt/edit instruction
Multi-frame HDR	Bright skies retained, shadows lifted, local contrast halos, compressed dynamic range.	Flattened shadows are not AI by themselves.	Use natural HDR only if phone-photo realism is intended; keep contact shadows.
Noise reduction smearing	Skin, walls, dark clothing, and hair edges lose fine texture.	Can mimic AI plastic skin or melted detail.	Restore microtexture selectively; do not over-sharpen faces.
Edge enhancement	Thin bright/dark halos around high-contrast edges.	Can resemble cutout halos.	Reduce halos or keep subtle phone-sharpened edge style if user wants phone realism.
Semantic bokeh	Portrait mode blur may fail around hair, glasses, fingers, and transparent objects.	Can look like AI boundary failure but may be real computational photography.	Correct depth mask transitions around hair, hands, and shoulders.
Small-sensor perspective	Wide lens, close subject, facial distortion near edges, deep depth of field.	Do not expect full-frame 85mm bokeh from casual phone images.	For phone realism: 24-28mm equivalent, high DOF, mild barrel distortion, HDR tone mapping.
Night mode	Brightened night, low noise, ghosting or motion artifacts, flattened shadows.	Clean low-light phone images are not automatically AI.	Use slight hand-held blur, realistic highlight clipping/halation, and plausible shadow noise.

Section 4 - Depth of Field, Focus, and Bokeh Physics

Depth of field depends on focal length, aperture, subject distance, sensor format, and viewing size. AI often applies blur decoratively rather than by distance from the focus plane. Phone portrait mode can also create mask-based bokeh that fails at boundaries.

Cue	Real physics	AI/composite failure	Enhancement rule
Focus plane	Objects at similar distance should share focus.	Same-distance objects randomly blur or sharpen.	Correct focus consistency before adding grain or colour grade.
Bokeh size	Background blur grows with distance from focus and wide aperture.	Blur amount ignores depth order.	Use realistic DOF based on camera/lens scenario.
Cat-eye bokeh	Fast lenses can show elliptical highlights near frame edges.	Random bokeh shapes without optical logic.	Only add if wide-aperture lens look is requested.
Portrait-mode blur	Phone software may blur around subject mask and fail at hair/glasses.	Could be real phone artifact or AI boundary failure.	Classify carefully; repair boundary masks when enhancing.

Section 5 - Light, Exposure, Tone, and Colour

Real exposure is a balance between light intensity, aperture, shutter speed, ISO, dynamic range, and processing decisions. Editing tools can radically alter tone while preserving real structure.

Concept	Meaning	Analysis instruction	Editing instruction
Inverse-square falloff	Light intensity from a point source drops with distance.	Check whether faces/objects at different distances have plausible brightness.	Match subject lighting to background before colour grading.
Dynamic range	Scenes can exceed sensor range; highlights clip or shadows block unless HDR/tone mapping is used.	No clipping is not suspicious in phone HDR; excessive flatness suggests processing.	Preserve some tonal anchors: true blacks, controlled highlights, midtone detail.
Colour temperature	Different light sources have different Kelvin values and spectral qualities.	Mixed lighting is normal indoors/streets; impossible mixed lighting is not.	Match white balance to environment; use local colour casts from windows/neon/sun.
Skin tone under light	Skin colour changes with light temperature, exposure, and reflections.	Do not confuse colour grade with complexion change.	Protect complexion identity; adjust luminance/chroma subtly and reference-based.
Clipping/bloom	Highlights can clip, bloom, halate, or reflect depending on	Modern HDR can suppress clipping; AI can simulate clipping.	Use clipping only where brightness warrants it.

	lens/sensor/material.		

Section 6 - Shadows, Contact Shadows, Bounce Light, and Ambient Occlusion

Grounding is one of the strongest realism factors. Contact shadows and ambient occlusion tell the viewer that an object occupies space. Composites and AI images often fail here.

Cue	What to check	Correction instruction
Contact shadow	Does each foot, object, wheel, chair leg, or garment hem touch the surface with a small darkening at contact?	Add subtle contact shadow under subject and objects; match softness to light size/distance.
Cast shadow	Does the shadow direction match the main light and other scene objects?	Do not add generic drop shadow; match angle, length, softness, and colour.
Ambient occlusion	Corners, underarms, neck under chin, folds, pockets, zippers, shoes, and object intersections should have local darkening.	Restore small occlusion shadows in folds and contact points.
Bounce light	Nearby surfaces tint shadows and skin/fabric with subtle colour.	Add environmental colour cast subtly; avoid flat grey shadows.

Section 7 - JPEG, Social Media, Screenshots, and Upscaling

Digital artifact	Appearance	Diagnostic warning
JPEG blocks	8x8 blockiness, mosquito noise around edges/text, banding in gradients.	Can make real text look strange; do not treat as pseudo-text unless glyph shapes are impossible.
Social recompression	Reduced fine detail, over-sharpening, resized edges, colour shifts.	Authenticity confidence must be lower for social downloads.
Screenshot capture	UI scaling, display gamma, moire, resampling, missing EXIF.	Screenshot metadata says little about original image capture.
AI upscaling/denoise	Plastic surfaces, invented pores, edge ringing, repeated microtexture.	Could be real image enhanced by AI tools; classify as manipulated/edited if base structure is real.
Banding	Posterised sky/background gradients due to compression or low bit depth.	Banding is not AI evidence by itself.

Section 8 - Colour Management and E-Commerce Delivery

For product imagery, realism also means reliable colour. Apparel and e-commerce images must keep fabric colour and texture consistent across pages. sRGB delivery remains the practical default for web storefronts.

- Use sRGB for web/storefront exports unless a platform specifies otherwise.
- Avoid extreme saturation/vibrance that misrepresents garment colour.
- Preserve neutral greys/whites to prevent colour casts in product pages.
- For fabric, sharpen weave and stitching but avoid crunchy halos around seams and logos.
- Check product colour under consistent lighting before applying film-style grading.

Section 9 - Realism Enhancement Priority Order

1. Correct geometry, perspective, and alignment.
2. Match subject and background lighting direction.
3. Add or repair contact shadows and ambient occlusion.
4. Restore material-specific texture: skin, fabric, metal, glass, pavement.
5. Correct depth of field and focus consistency.
6. Apply colour balance and tonal realism.
7. Add only subtle lens/sensor/film imperfections if appropriate.
8. Export in the right colour space and resolution for use case.

Quick Reference Summary

- Presence of imperfections supports realism; absence does not prove AI.
- Phone photos often look computational: HDR, denoise, sharpen, segmentation bokeh.
- Light and contact shadows matter more than generic grain.
- Use subtle, context-specific imperfections; avoid fake realism filters.
- For product photos, prioritize colour accuracy, clarity, fabric structure, and clean e-commerce delivery.

Source Notes and Reliability Labels

This document gives operational image-analysis and image-editing guidance. It must not claim legal or forensic certainty without original files, metadata, provenance, and human review. Use probability language.

- **Internal AuthenticLens Document 1 v1.0:** Source baseline for the AI artifact lexicon: anatomical anomalies, text failures, pattern distortions, boundary failures, AI glow, material physics, structural coherence, and model signatures.
- **Internal AuthenticLens Document 2 v1.0:** Source baseline for photographic physics: optical aberrations, vignetting, distortion, flare, diffraction, sensor noise, film grain, depth of field, reflection physics, exposure, tone, and colour temperature.
- **Internal AuthenticLens Document 3 v1.0:** Source baseline for realism prompting: positive/negative realism vocabulary, film stock and camera references, scenario prompt formulas, and platform strategies.
- **Google AI Developers - Image generation with Gemini:** Current source note: Gemini supports conversational image generation/editing with text, images, or both. Nano Banana refers to Gemini 2.5 Flash Image using model gemini-2.5-flash-image. Nano Banana Pro refers to Gemini 3 Pro Image Preview

using model gemini-3-pro-image-preview. Google states generated images include SynthID watermarking.

- **Adobe Lightroom Help - Effects and Details:** Current source note: Texture can smooth or accentuate texture without changing colour/tonality; Clarity changes edge contrast; Dehaze changes haze and can affect perceived detail/colour; Vignette and Grain are stylistic effects; Detail controls Sharpening, Noise Reduction, and Color Noise Reduction.
- **C2PA Technical Specification:** Current source note: Content Credentials use signed manifests and assertions to record provenance and edits. Absence of C2PA data is not proof that an image is fake, and presence of metadata should be interpreted with context.
- **W3C / IEC sRGB references:** Current source note: sRGB is the default web colour space; for e-commerce and social delivery, unmanaged or non-sRGB exports can shift colour appearance across devices.
- **IPTC Photo Metadata Standard:** Current source note: IPTC metadata structures descriptive, location, rights, and administrative data for professional image workflows. Metadata is useful context but is not standalone authenticity proof.

Reliability labels used inside the document: Confirmed source, Common photography practice, Image-forensics heuristic, Operational guidance, Prompting heuristic, Model-behaviour observation.